

Software Design Document (SDD) – Sample

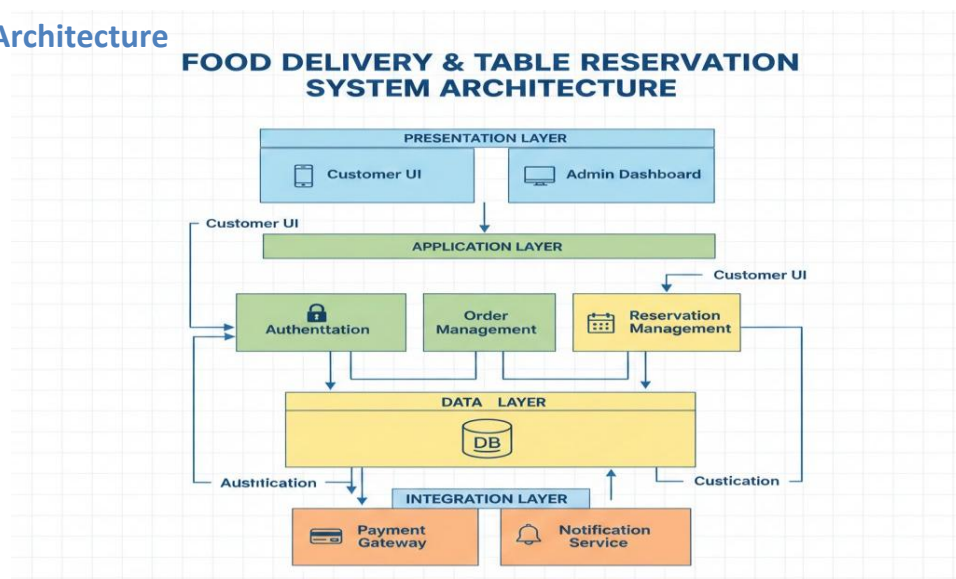
1. Introduction

The rapid growth of digital technology has significantly transformed the food service industry. Customers today expect quick access to restaurant services such as online food ordering, secure digital payments, and convenient table reservations. Traditional restaurant management systems that rely on manual processes often lead to delays, booking conflicts, and inefficient order handling.

The **Food Delivery & Table Reservation Application** is designed to address these challenges by providing an integrated web-based platform for customers and restaurant administrators. The system allows customers to browse menus, place food orders, make online payments, track order status, and reserve tables. At the same time, administrators can manage menus, view customer orders, update order statuses, and handle table reservations efficiently.

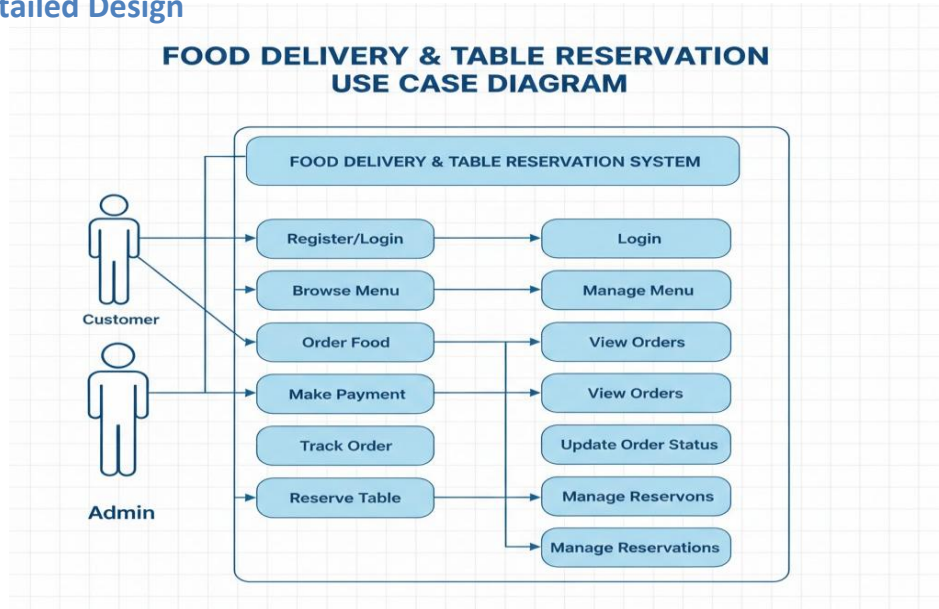
This application aims to improve customer convenience, reduce manual errors, and enhance overall restaurant operational efficiency through a secure and user-friendly digital solution.

2. System Architecture



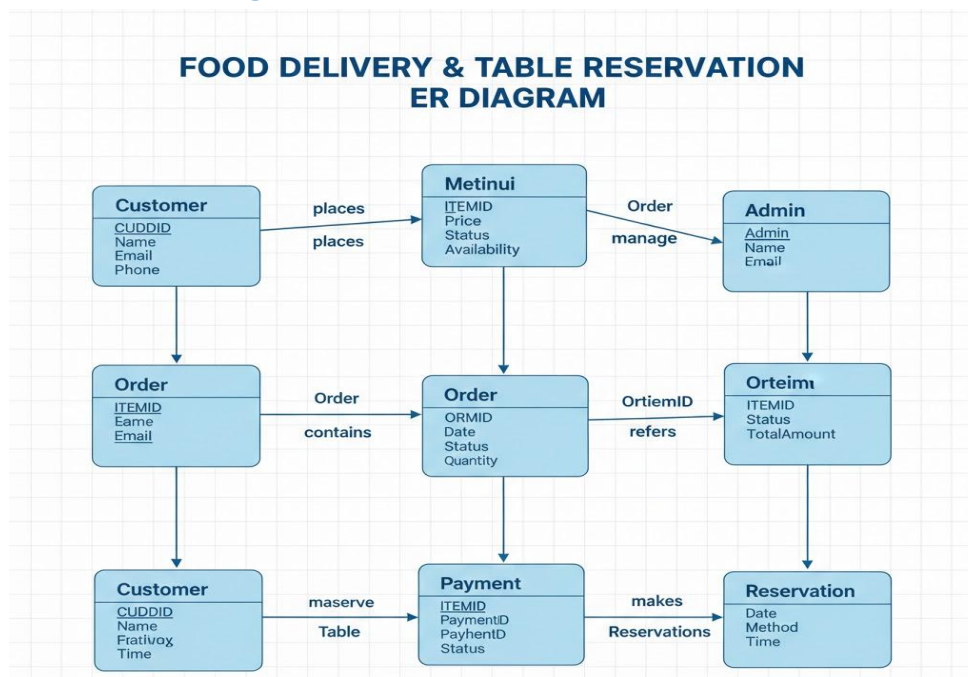
The system follows a layered architecture consisting of a presentation layer for user interaction, an application layer for business logic, a data layer for storage, and an integration layer for payment and notification services. This architecture ensures scalability, security, and maintainability.

3. Detailed Design



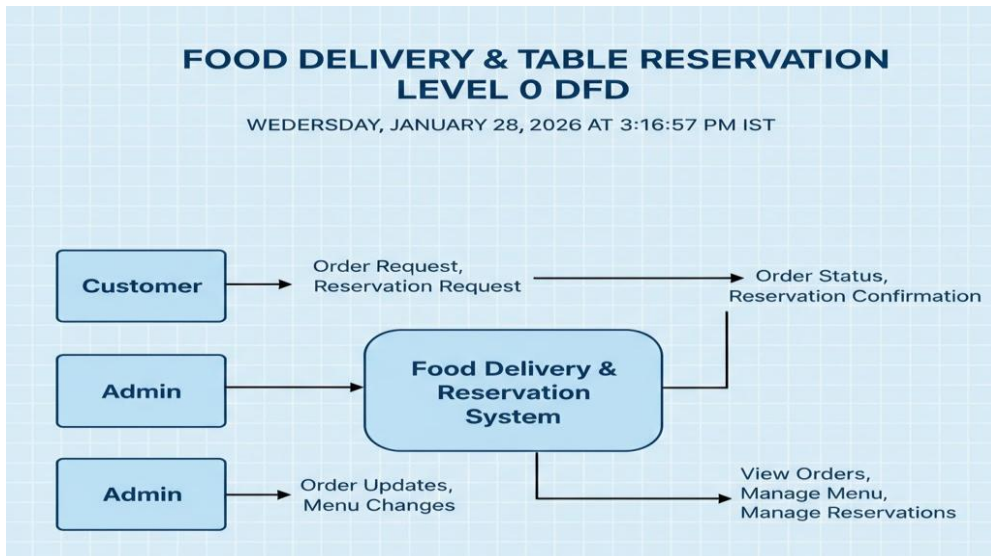
The use case diagram represents interactions between users and the system. Customers can browse menus, place orders, make payments, track orders, and reserve tables. Admins manage menus, view orders, update order statuses, and manage reservations.

4. Database Design



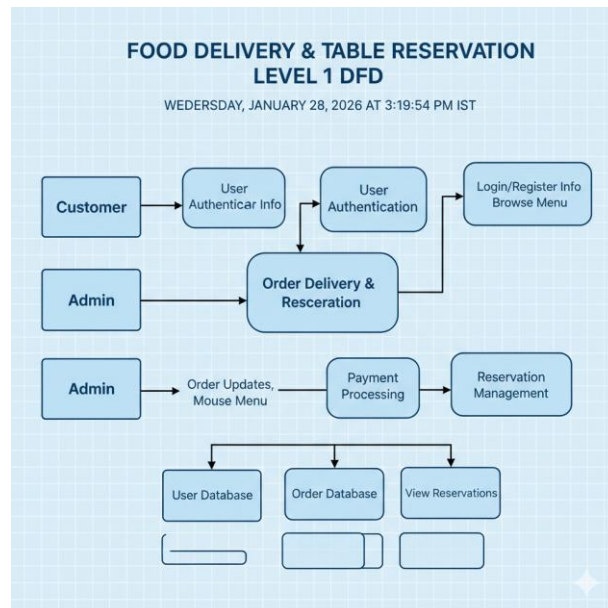
The ER diagram illustrates the database structure and relationships among entities such as Customer, Order, MenuItem, Payment, and Reservation. It ensures proper data organization and integrity.

4. DATA FLOW DIAGRAM (DFD - LEVEL 0)



The Level 1 DFD decomposes the system into sub-processes such as order processing, payment handling, and reservation management.

5. Design Constraints



This workflow diagram shows the step-by-step flow of food ordering from login to order delivery and status updates.